ECET
250E

Linear Circuits 1

Week 1 | Chapter 1: Basic Concepts (With Solutions)

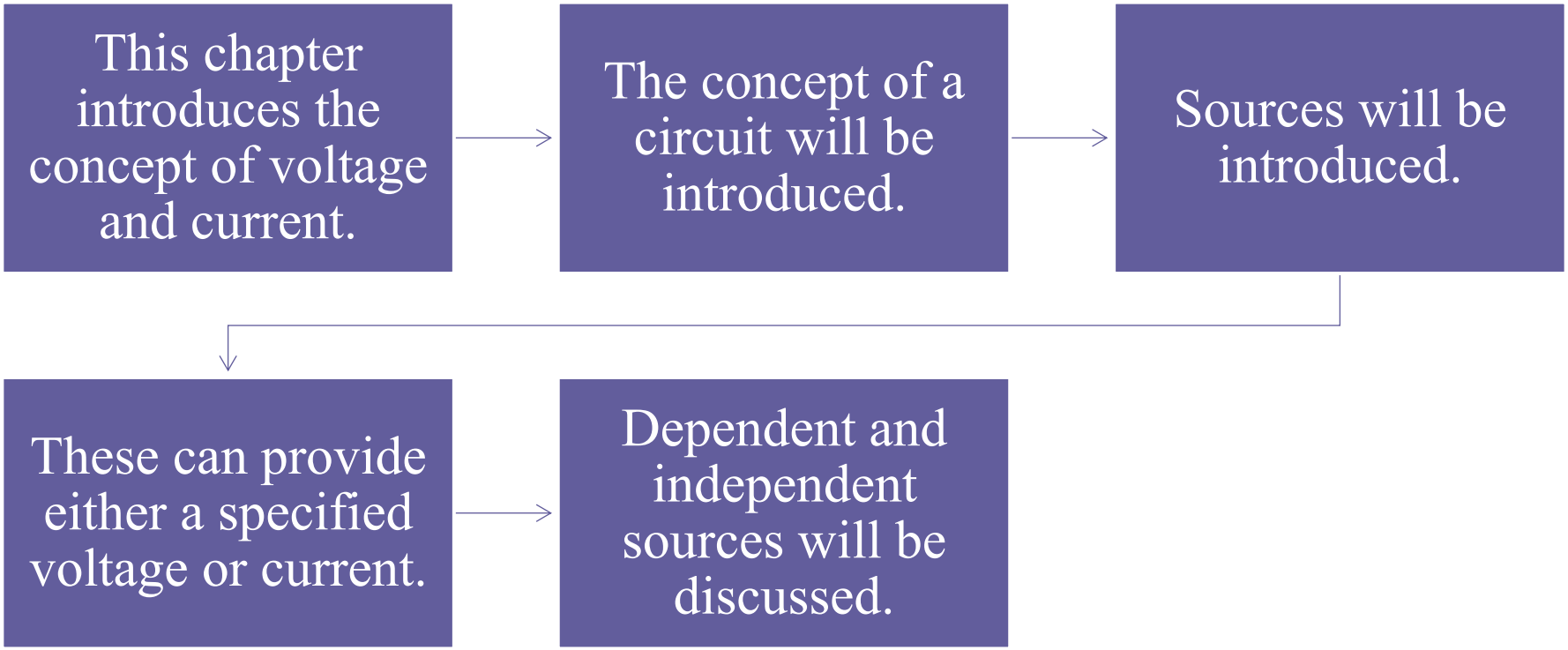
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Fall 2026

Based on Fundamentals of Electric Circuits, 2026 Release — C.K. Alexander, M.N.O. Sadiku & S.J. Fredette (McGraw Hill)

Introduction

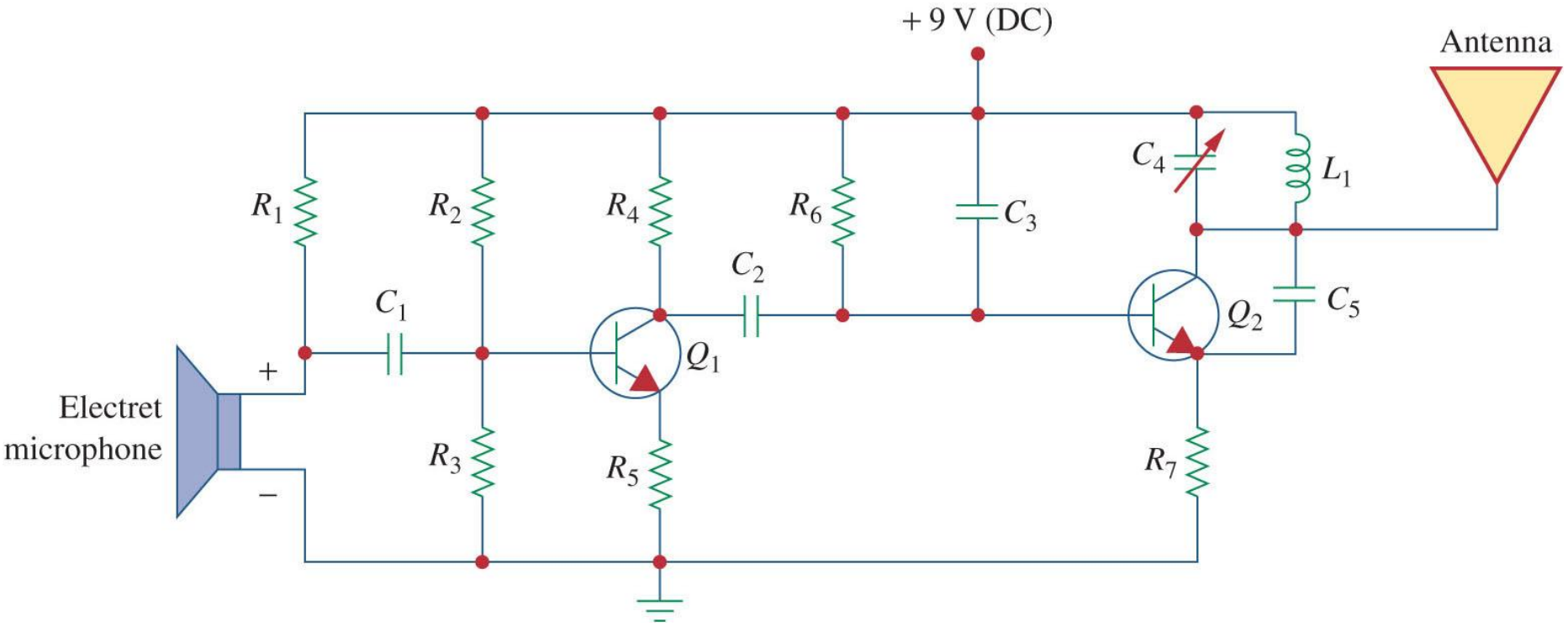


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What is a circuit?

- An electric circuit is an interconnection of electrical elements.
- It may consist of only two elements or many more:

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Units

- When taking measurements, we must use units to quantify values.
- We use the International Systems of Units (SI for short).
- Prefixes on SI units allow for easy relationships between large and small values.

TABLE 1.1

Six basic SI units and one derived unit relevant to this text.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

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TABLE 1.2

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

- In the lab, one typically sees (pC, nC, or μC).

Charge

- Charge is a basic SI unit, measured in Coulombs (C).
- Counts the number of electrons (or positive charges) present.
- Charge of single electron is $-1.602 * 10^{-19}$C.

- One Coulomb is quite large, $6.24*10^{18}$ electrons.

Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).

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Charge II

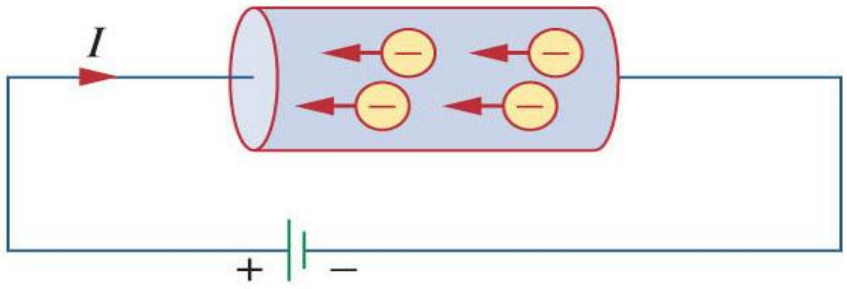
- Charge is always multiple of electron charge.
- *Charge cannot be created or destroyed, only transferred.*

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Current

- The movement of charge is called a current.
- Historically the moving charges were thought to be positive.
- Thus, we always note the direction of the equivalent positive charges, even if the moving charges are negative.

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Battery

A convention is a standard way of describing something so that others in the profession can understand what we mean. We will be using IEEE conventions throughout this book.

Current II

- Current, i, is measured as charge moved per unit time through an element.

$$i \equiv \frac{dq}{dt}$$

- Unit is Ampere (A), is one Coulomb/second.

1 ampere = 1 coulomb/second

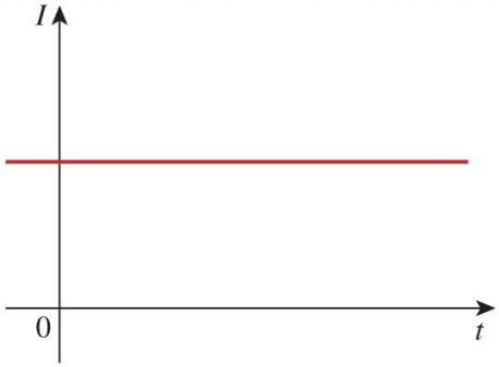
$$Q \triangleq \int_{t_0}^{t} i\,dt$$

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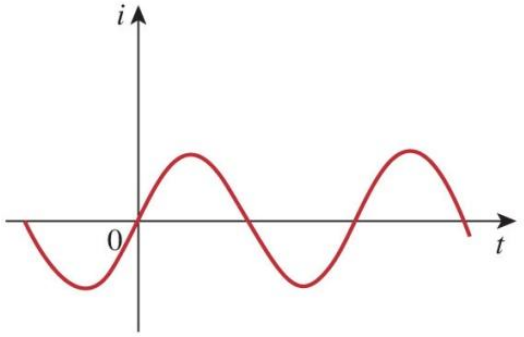
DC versus AC

- A current that remains constant with time is called Direct Current (DC).
- Such current is represented by the capital *I*, time varying current uses the lowercase, *i*.
- A common source of DC is a battery.
- A current that varies sinusoidally with time is called Alternating Current (A C).
- Mains power is an example of AC.

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(a)



(b)

A direct current (dc) flows only in one direction and can be constant or time varying.

An alternating current (ac) is a current that changes direction with respect to time.

Direction of Current

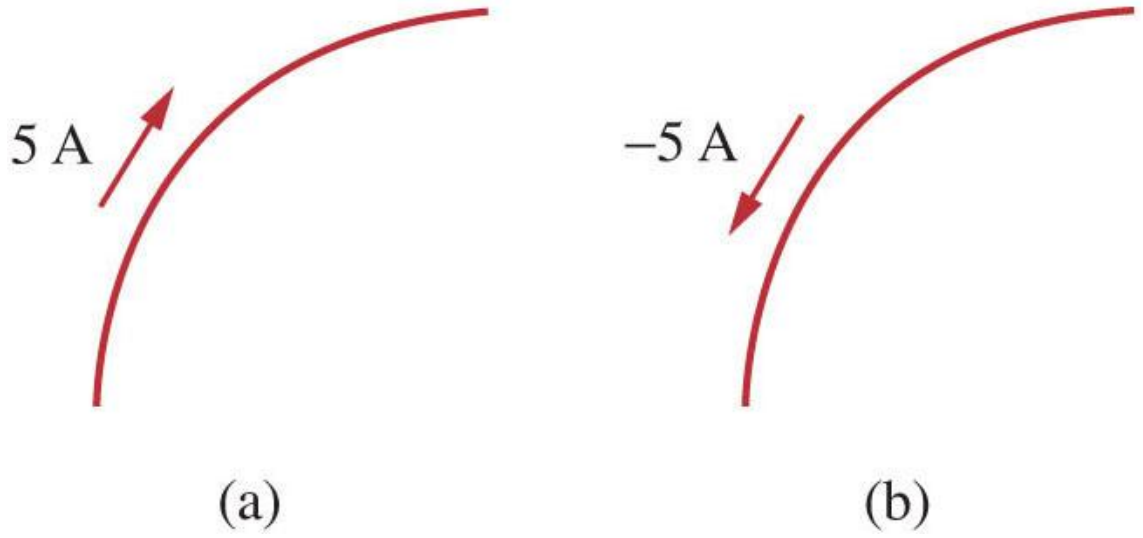
- The sign of the current indicates the direction in which the charge is moving with reference to the direction of interest we define.
- We need not use the direction that the charge moves in as our reference and often have no choice in the matter.

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Direction of Current II

- A positive current through a component is the same as a negative current flowing in the opposite direction.

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Examples

Example 1.1

How much charge is represented by 4,600 electrons?

Practice Problem 1.1

Calculate the amount of charge represented by 10 billion protons.

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Solutions

Example 1.1

SOLUTION

How much charge is represented by 4,600 electrons?

$Q = qe \times n = (1.602 \times 10^{-19}\ C)(4{,}600)$

$Q = -7.369 \times 10^{-16}\ C$ (negative — electron charge is negative)

Practice Problem 1.1

SOLUTION

Calculate the amount of charge represented by 10 billion protons.

$Q = qp \times n = (1.602 \times 10^{-19}\ C)(10 \times 10^{9})$

$Q = 1.6021 \times 10^{-9}\ C$

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Examples

Example 1.2

The total charge entering a terminal is given by

$$q(t) = 5t \sin(4\pi t) \text{ mC}.$$

Calculate the current at $t = 0.5$ s.

Practice Problem 1.2

If in Example 1.2,

$$q(t) = \left(20 - 15t - 10e^{-3t}\right) \text{ mC},$$

find the current at $t = 1.0$ s.

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Solutions

Example 1.2

SOLUTION

$q(t) = 5t \sin(4\pi t)$ mC. Find i at t = 0.5 s.

$i = dq/dt = 5 \sin(4\pi t) + 20\pi t \cos(4\pi t)$ mA

$i(0.5) = 5 \sin(2\pi) + 10\pi \cos(2\pi) = 0 + 10\pi$

i = 31.42 mA

Practice Problem 1.2

SOLUTION

$q(t) = (20 - 15t - 10e^{-3t})$ mC. Find i at t = 1.0 s.

$i = dq/dt = -15 + 30e^{-3t}$ mA

$i(1.0) = -15 + 30e^{-3} = -15 + 1.494$

i = −13.51 mA

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Examples

Example 1.3

Determine the total charge entering a terminal between

$$t = 1\,\text{s} \quad \text{and} \quad t = 2\,\text{s}$$

if the current passing the terminal is

$$i = (3t^2 - t)\,\text{A}$$

Practice Problem 1.3

The current flowing through an element is

$$i = \begin{cases} 8\,\text{A}, & 0 < t < 1 \\ 8t^2\,\text{A}, & t > 1 \end{cases}$$

Calculate the charge entering the element from

$$t = 0 \quad \text{to} \quad t = 2\,\text{s}$$

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Solutions

Example 1.3

SOLUTION

$i = (3t^2 - t)$ A. Find charge entering between $t = 1$ s and $t = 2$ s.

$Q = \int_1^2 (3t^2 - t)\, dt = [t^3 - t^2/2]_1^2 = (8 - 2) - (1 - 0.5)$

$Q = 5.5$ C

Practice Problem 1.3

SOLUTION

$i = 8$ A for $0 < t < 1$; $i = 8t^2$ A for $t > 1$. Find charge from $t = 0$ to $t = 2$ s.

$q = \int_0^1 8\, dt + \int_1^2 8t^2\, dt = 8 + (8/3)(8 - 1)$

$q = 26.67$ C

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Voltage

- Electrons move when there is a difference in charge between two locations.
- This difference is expressed at the potential difference, or voltage (V).
- It is always expressed with reference to two locations.

$$v_{ab} \triangleq \frac{dw}{dq}$$

$$1 \text{ volt} = 1 \text{ joule/coulomb} = 1 \text{ newton-meter/coulomb}$$

Voltage (or potential difference) is the energy required to move a unit charge from a reference point (−) to another point (+), measured in volts (V).

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Voltage II

- It is equal to the energy needed to move a unit charge between the locations.
- Positive charge moving from a higher potential to a lower yields energy.
- Moving from negative to positive requires energy.

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Voltage III

$v_{ab} = -v_{ba}$

For example, in Fig. 1.7, we have two representations of the same voltage. In Fig. 1.7(a), point a is +9 V above point b; in Fig. 1.7(b), point b is −9 V above point a. We may say that in Fig. 1.7(a), there is a 9-V voltage drop from a to b or equivalently a 9-V voltage rise from b to a. In other words, a voltage drop from a to b is equivalent to a voltage rise from b to a.

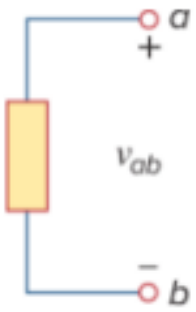
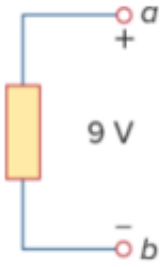
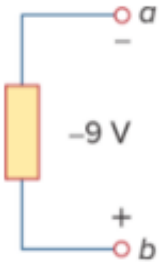


Figure 1.6

Polarity of voltage v_{ab}.



(a)



(b)

Figure 1.7

Two equivalent representations of the same voltage v_{ab}: (a) Point *a* is 9 V above point *b*; (b) point *b* is −9 V above point *a*.

Keep in mind that electric current is always *through* an element and that electric voltage is always *across* the element or between two points.

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Power and Energy

- Voltage alone does not equal power.
- It requires the movement of charge, *that is,* a current.
- Power is the product of voltage and current.

$$p = vi$$

$$p \triangleq \frac{dw}{dt}$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

- It is equal to the rate of energy provided or consumed per unit time.
- It is measured in Watts (W).

1) $v = \frac{dw}{dq}$

2) $i = \frac{dq}{dt}$

3) $p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$

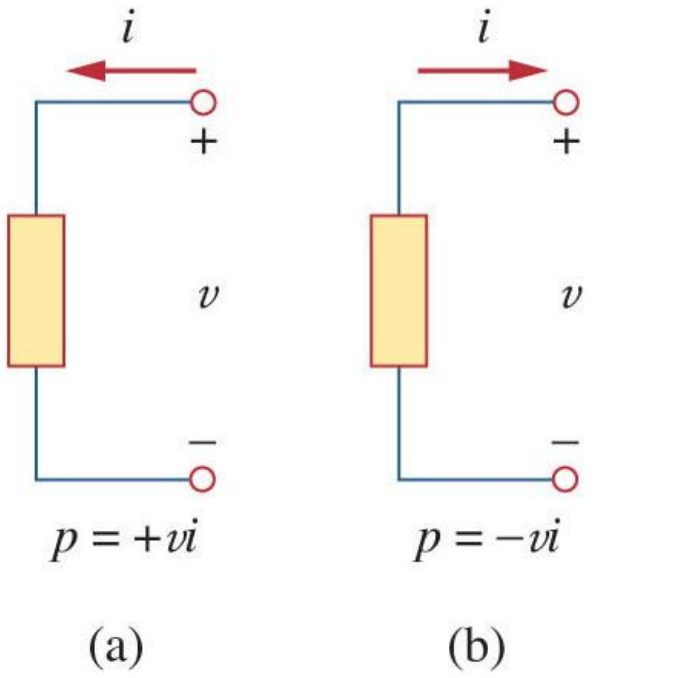
Power is the time rate of expending or absorbing energy, measured in watts (W).

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Passive Sign Convention

- By convention, we say that an element being supplied power has positive power.
- A power source, such as a battery has negative power.
- Passive sign convention is satisfied if the direction of current is selected such that current enters through the terminal that is more positively biased.

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When the voltage and current directions conform to Fig. 1.8(b), we have the *active sign convention* and $p = +vi$.

Passive sign convention is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$.

Conservation of Energy

- In a circuit, energy cannot be created or destroyed.
- Thus power also must be conserved.
- The sum of all power supplied must be absorbed by the other elements.
- Energy can be described as watts x time.
- Power companies usually measure energy in watt-hours.

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Power and Energy

+Power absorbed = − Power supplied



Figure 1.9

Two cases of an element with an absorbing power of 12 W: (a) $p = 4 \times 3 = 12$ W, (b) $p = 4 \times 3 = 12$ W.



Figure 1.10

Two cases of an element with a supplying power of 12 W: (a) $p = -4 \times 3 = -12$ W, (b) $p = -4 \times 3 = -12$ W.

In fact, the **law of conservation of energy** must be obeyed in any electric circuit. For this reason, the algebraic sum of power in a circuit, at any instant of time, must be zero:

$\sum p = 0$

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Power and Energy

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

Energy is the capacity to do work, measured in joules (J).

The electric power utility companies measure energy in watt-hours (Wh), where

$$1 \text{ Wh} = 3{,}600 \text{ J}$$

Example 1.4

An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

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Solutions

Example 1.4

SOLUTION

Constant current i = 2 A for t = 10 s; light + heat energy W = 2.3 kJ.

Total charge: dq = i × t = (2 A)(10 s) = 20 C

Voltage drop: v = dW/dq = 2,300 J / 20 C = 115 V

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Power and Energy Examples

Practice Problem 1.4

To move charge q from point b to point a requires 100 J.

Find the voltage drop v_{ab} (the voltage at a positive with respect to b) if:

(a) $q = 5\,\mathrm{C}$

(b) $q = -10\,\mathrm{C}$

Example 1.5

Find the power delivered to an element at $t = 3\,\mathrm{ms}$ if the current entering its positive terminal is:

$$i(t) = 5\cos(60\pi t)\,\mathrm{A}$$

and the voltage is:

(a) $v = 3i$

(b) $v = 3\frac{di}{dt}$

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Solutions

Practice Problem 1.4

SOLUTION

Moving q from b to a takes 100 J. Find $v_a^b = dW/dq$.

(a) q = 5 C: $v_a^b = 100\ \text{J} / 5\ \text{C} = 20\ \text{V}$

(b) q = −10 C: $v_a^b = 100\ \text{J} / -10\ \text{C} = -10\ \text{V}$

Example 1.5

SOLUTION

(a) $v = 3i$

$$p = vi = (3i)i = 3i^2$$
$$p(t) = 3[5cos(60\pi t)]^2 = 75cos^2(60\pi t)$$
$$p(3\ \text{ms}) = 75cos^2(60\pi \times 0.003) = 53.48\ \text{W}$$

(b) $v = 3\frac{di}{dt}$

$$\frac{di}{dt} = 5(-60\pi)sin(60\pi t) = -300\pi sin(60\pi t)$$
$$v = 3\frac{di}{dt} = -900\pi sin(60\pi t)$$
$$p = vi = [-900\pi sin(60\pi t)][5cos(60\pi t)] = -4500\pi sin(60\pi t)cos(60\pi t)$$
$$p(3\ \text{ms}) = -4500\pi sin(0.18\pi)cos(0.18\pi) = -6.40\ \text{kW}$$

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Power and Energy Examples

Practice Problem 1.5

Find the power delivered to the element in [Example 1.5] at

$$t = 5\,\text{ms}$$

if the current remains the same but the voltage is:

(a) $v = 6i\,\text{V}$

(b) $v = \left(6 + 10\int_0^t i\,dt\right)\,\text{V}$

Example 1.6

How much energy does a 100-W electric bulb consume in two hours?

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Solutions

Practice Problem 1.5

SOLUTION

$$i(t) = 5cos(60\pi t)\ \text{A}, t = 5\ \text{ms} = 0.005\ \text{s}$$

(a) $v = 6i$

$$p = vi = 6i^2 = 6[5cos(60\pi t)]^2 = 150cos^2(60\pi t)$$
$$p(0.005) = 150cos^2(0.3\pi) = 51.82\ \text{W}$$

(b) $v = 6 + 10\int_0^t i\,(\tau)\,d\tau$

$$\int_0^t i\,(\tau)\,d\tau = \int_0^t 5\,cos(60\pi\tau)\,d\tau = \left[\frac{5}{60\pi}sin(60\pi\tau)\right]_0^t = \frac{sin(60\pi t)}{12\pi}$$
$$v(t) = 6 + \frac{10}{12\pi}sin(60\pi t)$$
$$v(0.005) = 6 + \frac{10}{12\pi}sin(0.3\pi) = 6.2146\ \text{V}$$
$$i(0.005) = 5cos(0.3\pi) = 2.9389\ \text{A}$$
$$p = vi = (6.2146)(2.9389) = 18.26\ \text{W}$$

Example 1.6

SOLUTION

How much energy does a 100-W bulb consume in 2 hours?

W = P × t = 100 W × 2 h = 200 Wh

W = 720,000 J = 720 kJ

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Power and Energy Examples

Practice Problem 1.6

A home electric heater draws 12 A when connected to a 115 V outlet. How much energy is consumed by the heater over a period of 24 hours?

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Solutions

Practice Problem 1.6 | **SOLUTION**

Heater draws 12 A at 115 V. Find energy consumed over 24 hours.

$P = V \times I = 115\ V \times 12\ A = 1{,}380\ W$

$W = P \times t = 1{,}380\ W \times 24\ h = 33{,}120\ Wh$

$W = 33.12\ kWh$

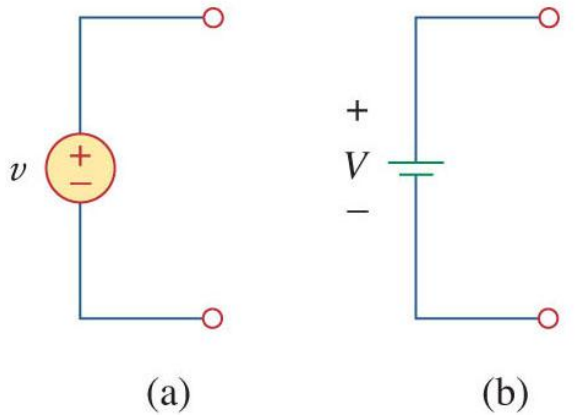
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Circuit Elements

Two types:

- Active: Active elements can generate energy.
- Passive.

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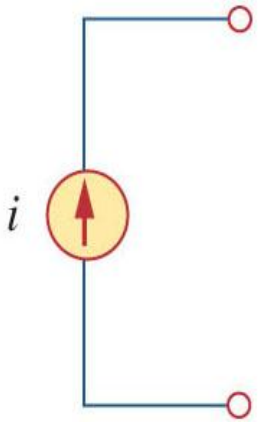


The most important active elements are voltage and current sources.

Active elements examples:

- Generators.
- Batteries.
- Operational Amplifiers.

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Circuit Elements II

Passives absorb energy.

- Resistors.
- Capacitors.
- Inductors.

But it should be noted that only the resistor dissipates energy ideally.

The inductor and capacitor do not.

An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

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Ideal Voltage Source

- An ideal voltage source has no internal resistance.
- It also is capable of producing any amount of current needed to establish the desired voltage at its terminals.
- Thus, we can know the voltage at its terminals, but we don't know in advance the current.

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Ideal Current Source

- Current sources are the opposite of the voltage source:
- They have infinite resistance.
- They will generate any voltage to establish the desired current through them.
- We can know the current through them in advance, but not the voltage.

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Ideal sources

- Both the voltage and current source ideally can generate infinite power.
- They are also capable of absorbing power from the circuit.
- It is important to remember that these sources do have limits in reality:
- Voltage sources have an upper current limit.
- Current sources have an upper voltage limit.

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Dependent Sources

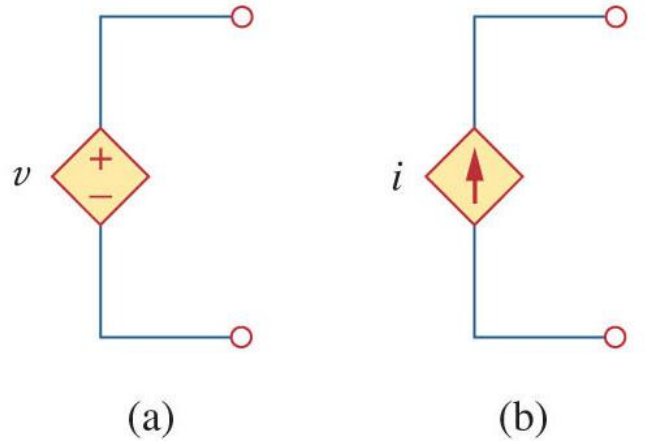
A dependent source has its output controlled by an input value.

Symbolically represented as a diamond.

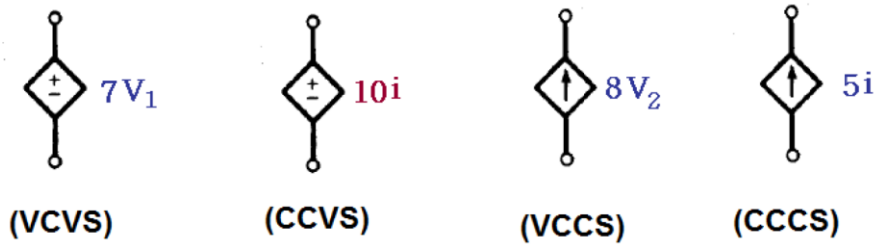
Four types:

- A voltage-controlled voltage source (VCVS).
- A current-controlled voltage source (CCVS).
- A voltage-controlled current source (VCCS).
- A current-controlled current source (CCCS).

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Which type of source is represented by the figures below.

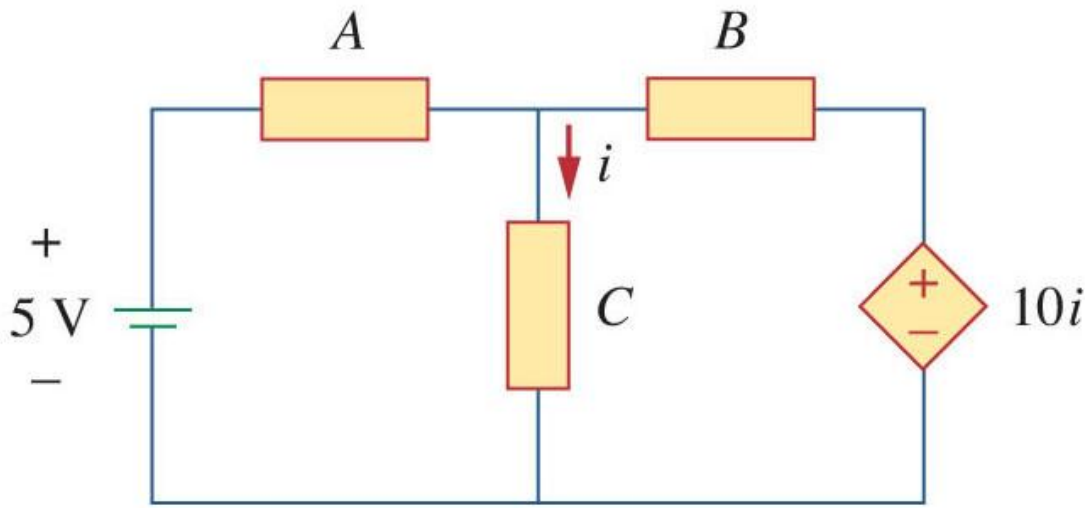


An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

Dependent Source example

- The circuit shown below is an example of using a dependent source.
- The source on the right is controlled by the current passing through element C.

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Circuit Elements Example

Example 1.7

Calculate the power supplied or absorbed by each element in Fig. 1.15.

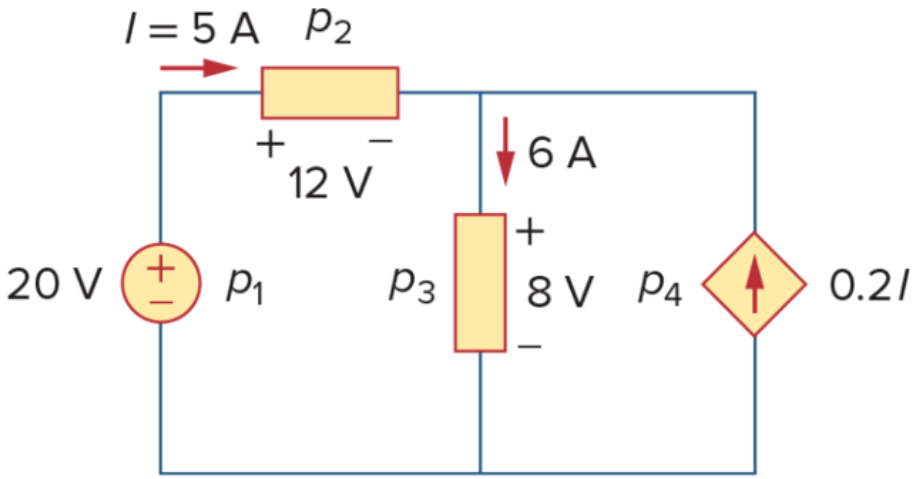


Figure 1.15
For Example 1.7.

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Circuit Elements Example

Example 1.7 — Solution

SOLUTION

$P_1 = 20\ V \times (-5\ A) = -100\ W$ supplying power

$P_2 = 12\ V \times 5\ A = 60\ W$ absorbing power

$P_3 = 8\ V \times 6\ A = 48\ W$ absorbing power

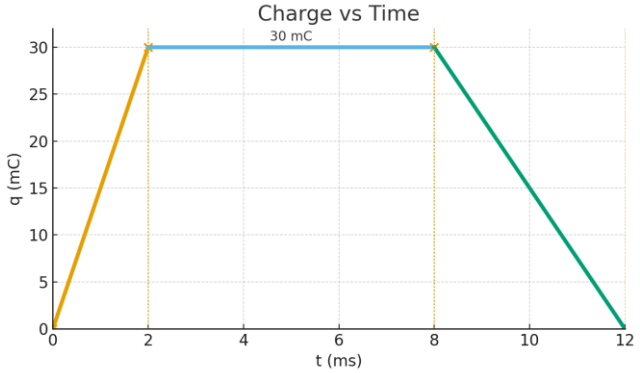
$P_4 = 8\ V \times (-0.2 \times 5\ A) = 8\ V \times (-1\ A) = -8\ W$ supplying power

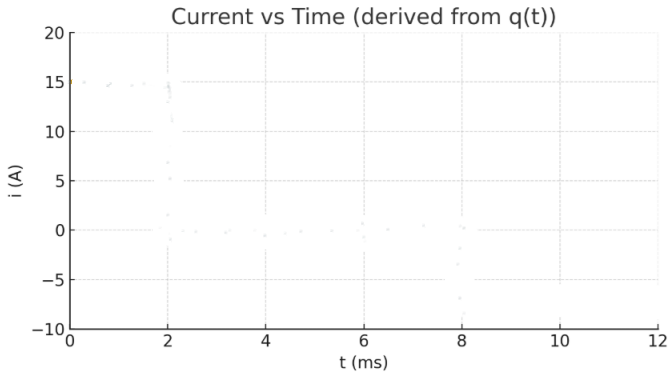
Check: $P_1+P_2+P_3+P_4 = -100+60+48-8 = 0$ (conservation of energy)

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Extra Examples for charge and current

The charge entering a certain element is shown in the figure below.
Find the current at:
(a) t = 2 ms **(b)** t = 6 ms **(c)** t = 8 ms
(Assume q(t) is in mC and t is in ms.) and Sketch the corresponding current.





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Extra Example — Solution

SOLUTION

i(t) = dq(t)/dt, read from the slope of the q(t) graph.

(a) t = 2 ms linear rise, 0→30 mC over 0–2=2 ms:

i = 30 mC / 2 ms = 15 mA

(b) t = 6 ms flat segment (q constant at 30 mC):

i = 0 mA

(c) t = 8 ms start of linear fall, 30→0 mC over 8–12=4 ms:

i = (0 − 30) mC / 4 ms = −7.5 mA

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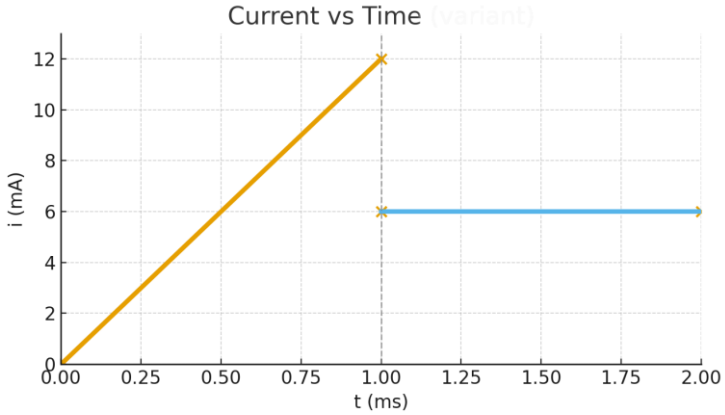
Example

The current flowing past a point in a device is shown in Fig. P1.8-variant.

- From 0 to $1\,\text{ms}$, $i(t)$ increases linearly from 0 to $12\,\text{mA}$.
- At $t = 1\,\text{ms}$ the current steps down.
- From 1 to $2\,\text{ms}$, $i(t) = 6\,\text{mA}$ (constant).

Calculate the total charge that passes the point over $0 \le t \le 2\,\text{ms}$.

(Use $q = \int i\,dt$; note $1\,\text{mA}\cdot\text{ms} = 1\,\mu\text{C}$.)



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Solution

Charge from a Current Graph | **SOLUTION**

$q = \int i\,dt$ = area under i(t).

Triangle, 0 → 1 ms (0 to 12 mA):

$q_1 = \frac{1}{2}(1\text{ ms})(12\text{ mA}) = 6\ \mu C$

Rectangle, 1 → 2 ms (constant 6 mA):

$q_2 = (6\text{ mA})(1\text{ ms}) = 6\ \mu C$

Total charge:

$q = q_1 + q_2 = 12\ \mu C$

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